

Skills Summary

Halves and Fourths That Really Are Equal

Use this reference while completing your worksheet or when studying.

Skill 1 — Fixing two parts that are called halves

Rule: *Halves* means **two parts of the same size**. Two pieces alone are not enough — they must match.

To check a strip of equal squares: count the squares in each part. If the counts differ, the parts are not halves.

To fix it: share all the squares equally between the two parts, each half = whole $\div$ two.



Example: A strip of 10 squares is cut into a part of 7 squares and a part of 3. Are they halves? If not, fix them.

Step 1. Halves have to be the same size, so before anything else I compare the two counts instead of trusting the label.

Step 2. Seven squares against three squares is not a match, so these are not halves. Sharing all ten squares equally between two parts gives each part $10 \div 2$ squares, so the cut belongs two squares further along.

not halves: $7 > 3$ each true half: 5 squares

Try it: A strip of 6 squares is cut into a part of 4 and a part of 2. Are they halves? How many squares should each half have?

check: not halves, $4 > 2$; each true half 3 squares

Skill 2 — Fixing four parts that are called fourths

Rule: *Fourths* means **four parts of the same size**. Four pieces alone are not enough — all four must match.

Check the biggest part against the smallest: if those two differ, the parts are not fourths.

To fix it: each fourth = whole $\div$ four.

Example: A strip of 12 squares is cut into parts of 6, 2, 2 and 2 squares, and called fourths. Are they?

Step 1. All four parts have to be the same size, so the quickest test is to set the biggest part against the smallest and see whether even those two match.

Step 2. Six squares against two squares is not a match, so these are not fourths. Sharing all twelve squares among four equal parts gives $12 \div 4$ squares each, so the big part must give squares back to the others.

not fourths: $6 > 2$ each true fourth: 3 squares

Try it: A strip of 8 squares is cut into parts of 5, 1, 1 and 1, and called fourths. Are they? How many squares should each fourth have?

check: not fourths, 5 > 1; each true fourth 2 squares

Skill 3 — Halves are bigger than fourths

Rule: The more equal parts a whole is shared into, the *smaller* each part gets — the whole never grows.

So a half of a strip is always bigger than a fourth of that same strip.

Example: A strip of 8 squares is shared into equal halves, and a strip just like it into equal fourths. How big is each part?

Step 1. Both strips hold the same amount, so the only thing that changes is how many equal parts it is shared into — that tells me to share the same total two ways and then four ways.

Step 2. Shared between two: $8 \div 2$ squares each. Shared among four: $8 \div 4$ squares each. The same strip, more parts, smaller parts.

each half: 4 squares each fourth: 2 squares

Try it: A strip of 16 squares is shared into equal halves, and another into equal fourths. How big is each part?

check: each half 8 squares, each fourth 4 squares